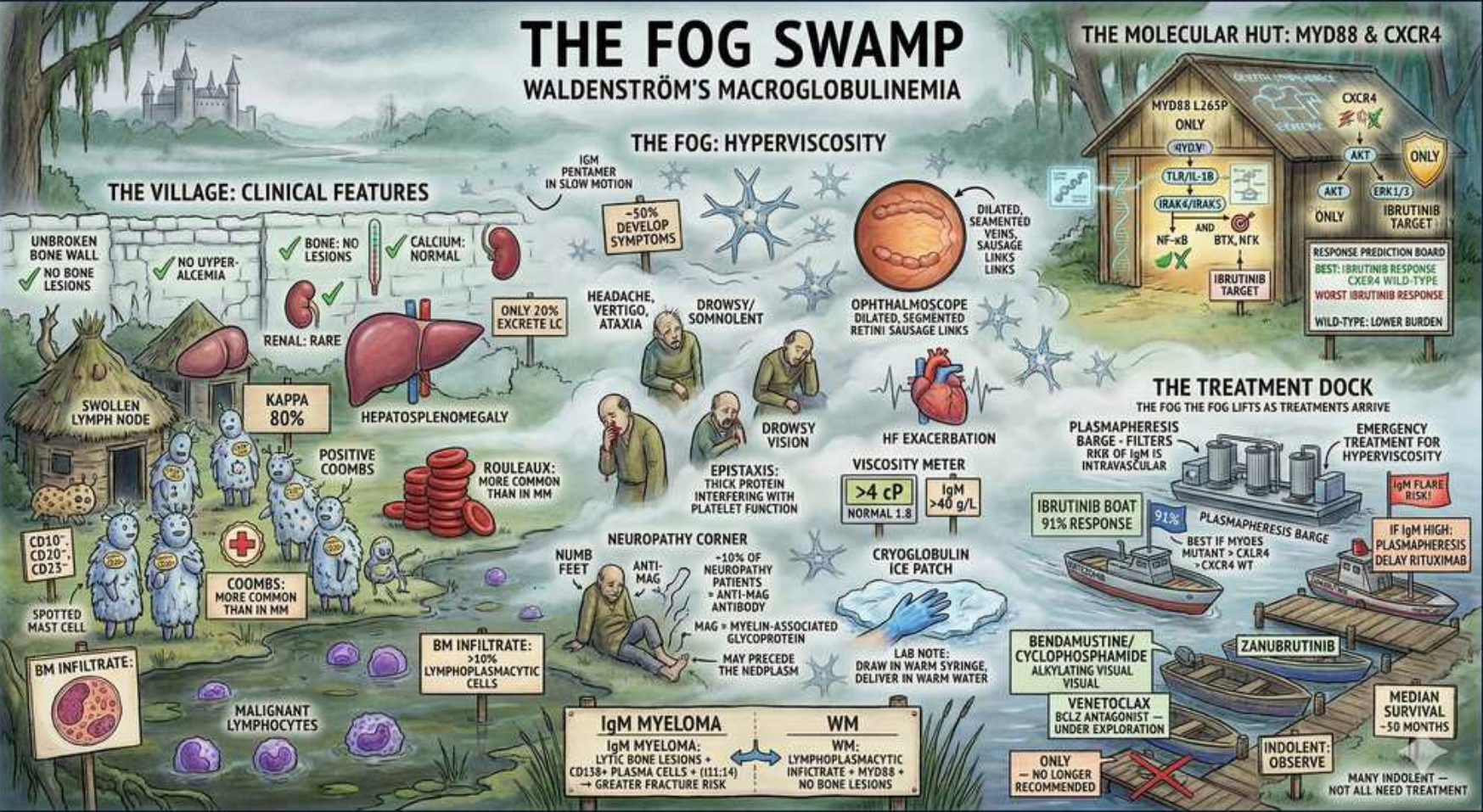


Plasma Cell — Waldenström Macroglobulinemia & Hyperviscosity



1. IgM monoclonal

WHAT YOU SEE

IgM 'fog' rolling through the swamp

WHAT IT ENCODES

Waldenström macroglobulinemia (WM) is a lymphoplasmacytic lymphoma whose clonal cells secrete a monoclonal IgM paraprotein — the largest immunoglobulin, a pentamer (~970 kDa) that stays intravascular and drives hyperviscosity. Diagnostic threshold is any detectable monoclonal IgM plus ≥10% clonal lymphoplasmacytic marrow infiltration; IgM MGUS is the precursor. Boards anchor: think IgM = 'I aM big' (pentamer) and 'fog in the swamp' = sludgy serum.

BOARD TRAP

WRONG answer: calling this 'IgM multiple myeloma.' WM is a low-grade B-cell/lymphoplasmacytic LYMPHOMA, not a plasma cell myeloma — discriminator: WM has lymphadenopathy/splenomegaly and NO lytic bone lesions or hypercalcemia, whereas myeloma has CRAB.

2. No bone lesions

WHAT YOU SEE

'No bone lesions, no hypercalcemia' village signs

WHAT IT ENCODES

Unlike multiple myeloma, WM characteristically spares the skeleton — no osteolytic lesions, no pathologic fractures, and hypercalcemia is rare because there is little osteoclast activation. The infiltrate is in marrow, nodes, and spleen rather than destroying cortical bone. Use absence of lytic lesions as a positive discriminator FOR WM against IgM myeloma.

BOARD TRAP

WRONG answer: ordering a skeletal survey expecting punched-out lesions to confirm WM. Lytic bone lesions or hypercalcemia point AWAY from WM and toward the rare IgM multiple myeloma — finding them should make you re-examine the diagnosis.

3. Hyperviscosity syndrome

WHAT YOU SEE

Fog = hyperviscosity, headache/ataxia/somnolence

WHAT IT ENCODES

The bulky pentameric IgM raises serum viscosity, producing the triad of mucosal/spontaneous bleeding, visual disturbance, and neurologic symptoms (headache, dizziness, ataxia, somnolence, confusion). Sludging in the microcirculation impairs perfusion; severe cases progress to stupor or coma. Mnemonic for hyperviscosity: 'bleeding, blurring, blundering' (bleeding, vision, neuro).

BOARD TRAP

WRONG answer: attributing the headache/confusion to brain metastases or stroke and ordering thrombolytics. The discriminator is the constellation with mucosal bleeding + retinal vein changes + elevated serum viscosity in an IgM patient — the fix is plasmapheresis, not tPA, and IgM (not IgG) is the usual culprit.

4. Sausage-link retina

WHAT YOU SEE

Ophthalmoscope showing dilated segmented retinal veins

WHAT IT ENCODES

Funduscopy shows dilated, tortuous, segmented retinal veins ('sausage-link' or 'boxcar' veins), often with flame hemorrhages and papilledema — a direct window onto microvascular sludging. Patients report blurred vision or visual loss. Dilated funduscopy is a fast bedside clue to symptomatic hyperviscosity.

BOARD TRAP

WRONG answer: diagnosing diabetic or hypertensive retinopathy and deferring treatment. The discriminator is the symmetric segmented 'boxcar' venous dilation appearing acutely alongside systemic hyperviscosity symptoms — it is an indication for urgent plasmapheresis, not routine ophthalmology follow-up.

5. Mucosal bleeding

WHAT YOU SEE

Epistaxis, IgM coating platelets

WHAT IT ENCODES

IgM coats platelets and binds/sequesters coagulation factors (and stretches vessel walls via sludging), producing mucosal bleeding such as epistaxis and gingival oozing despite a normal platelet count. This is an acquired qualitative platelet/coagulation defect, not thrombocytopenia. Bleeding is one leg of the hyperviscosity triad.

BOARD TRAP

WRONG answer: transfusing platelets or giving vitamin K for the epistaxis. The discriminator is that counts and INR are often near-normal — the defect is paraprotein-mediated platelet dysfunction, so the real fix is removing IgM with plasmapheresis.

6. Viscosity meter

WHAT YOU SEE

Viscosity meter, symptoms when >4 cP

WHAT IT ENCODES

Serum viscosity is reported relative to water (normal ~1.4–1.8 cP); symptoms usually emerge above ~4 cP and are common above 5–6 cP, but the threshold is patient-specific. Treat the patient, not the number — symptomatic hyperviscosity warrants plasmapheresis regardless of an exact value. Serum viscosity, not IgM level alone, is the functional measure.

BOARD TRAP

WRONG answer: withholding plasmapheresis because the measured viscosity is 'only' 3.5 cP in a clearly symptomatic bleeding/visual-change patient. The discriminator is clinical: there is no rigid numeric cutoff, so symptoms override a borderline viscosity reading.

7. Plasmapheresis first

WHAT YOU SEE

Plasmapheresis barge filtering IgM

WHAT IT ENCODES

Symptomatic hyperviscosity is an oncologic emergency: do urgent plasmapheresis, which rapidly removes the intravascular IgM (most IgM is intravascular, so apheresis is very effective) and reverses symptoms within hours. It is a bridge only — definitive control requires systemic therapy (e.g., a BTK inhibitor or rituximab-based regimen). Mnemonic: 'pull the plug on the fog' before chemo.

BOARD TRAP

WRONG answer: starting rituximab/chemotherapy first in a symptomatic hyperviscous patient. Plasmapheresis must come FIRST — discriminator: rituximab can transiently RAISE IgM ('IgM flare') and worsen viscosity, and chemo is too slow to relieve an acute emergency.

8. Hepatosplenomegaly/nodes

WHAT YOU SEE

Hepatosplenomegaly + lymph nodes in village

WHAT IT ENCODES

Hepatomegaly, splenomegaly, and lymphadenopathy reflect the underlying lymphoplasmacytic lymphoma infiltrating reticuloendothelial organs — features of a lymphoma, not of myeloma. They support WM over a plasma cell dyscrasia. B symptoms (fatigue, weight loss) may accompany.

BOARD TRAP

WRONG answer: interpreting splenomegaly + lymphadenopathy as evidence for multiple myeloma. The discriminator is that myeloma rarely causes organomegaly/adenopathy — their presence with an IgM spike favors WM (or another lymphoma).

9. Cold agglutinin / Coombs

WHAT YOU SEE

Positive Coombs, kappa 80%

WHAT IT ENCODES

Monoclonal IgM frequently acts as a cold agglutinin, causing complement-mediated autoimmune hemolytic anemia with a direct Coombs (DAT) positive for C3d; clones are usually kappa light chain. Cold-exposure worsens hemolysis and acrocyanosis. Cold AIHA = IgM + C3d on DAT (vs warm AIHA = IgG).

BOARD TRAP

WRONG answer: treating the hemolysis like warm AIHA with corticosteroids/splenectomy. The discriminator is the IgM/C3d cold pattern — steroids and splenectomy are largely ineffective; management is cold avoidance plus treating the WM (rituximab-based), and transfusions use a blood warmer.

10. Anti-MAG neuropathy

WHAT YOU SEE

Neuropathy corner, anti-MAG, numb feet

WHAT IT ENCODES

Monoclonal IgM directed against myelin-associated glycoprotein (anti-MAG) causes a distal, symmetric, predominantly sensory DEMYELINATING peripheral neuropathy with gait ataxia and numb feet. EMG shows demyelination; anti-MAG titers confirm. Classic exam image: chronic length-dependent sensory neuropathy in an IgM patient.

BOARD TRAP

WRONG answer: labeling the numb feet as diabetic neuropathy or simple CIDP and missing the paraprotein. The discriminator is the IgM anti-MAG antibody and demyelinating EMG pattern — check SPEP/immunofixation; treatment targets the IgM clone (e.g., rituximab), not just glycemic control.

11. Cryoglobulinemia

WHAT YOU SEE

Cryoglobulin ice patch

WHAT IT ENCODES

Monoclonal IgM can be a Type I cryoglobulin that precipitates in the cold, causing acrocyanosis, Raynaud-like digital ischemia, livedo, and a cold-induced vasculopathy. Type I cryos are single monoclonal Ig (vs mixed Type II/III with rheumatoid factor activity, often HCV-related). Lab pitfall: keep the sample warm (37°C) or you miss it.

BOARD TRAP

WRONG answer: classifying it as mixed (Type II) cryoglobulinemia and chasing hepatitis C. The discriminator is that WM produces a monoclonal Type I cryoglobulin (single IgM), presenting with occlusive cold-induced ischemia rather than the immune-complex small-vessel vasculitis/glomerulonephritis of mixed cryos.

12. MYD88 / CXCR4

WHAT YOU SEE

Molecular hut MYD88 L265P + CXCR4

WHAT IT ENCODES

The MYD88 L265P mutation is present in ~90–95% of WM and activates NF-κB downstream of BTK, making it a hallmark and a predictor of BTK-inhibitor response. CXCR4 (WHIM-like) mutations co-occur in ~30–40% and blunt/delay ibrutinib response and raise IgM. Mnemonic: MYD88 = 'My Drug 88' (ibrutinib works); CXCR4 = 'eXtra resistant.'

BOARD TRAP

WRONG answer: calling MYD88 L265P specific/diagnostic enough to skip the marrow biopsy, or expecting full ibrutinib response regardless of CXCR4. Discriminators: MYD88 L265P also occurs in some other lymphomas (not 100% specific), and CXCR4 mutation predicts a blunted, slower response to single-agent ibrutinib.

13. BTK inhibitor

WHAT YOU SEE

Ibrutinib/zanubrutinib boat in treatment dock

WHAT IT ENCODES

BTK inhibitors (ibrutinib, zanubrutinib) are highly active oral targeted agents in WM, exploiting MYD88-driven BTK signaling; zanubrutinib is more selective with less atrial fibrillation/bleeding than ibrutinib. Response is best with MYD88-mutated/CXCR4-wild-type disease. Watch for AF, bleeding, and hypertension with ibrutinib.

BOARD TRAP

WRONG answer: choosing a BTK inhibitor as monotherapy for a patient in acute hyperviscosity crisis. The discriminator is timing — plasmapheresis treats the emergency first; BTK inhibitors are durable disease control but do not act fast enough for a crisis.

14. Rituximab IgM flare

WHAT YOU SEE

Rituximab barge, delay if high IgM

WHAT IT ENCODES

Rituximab (anti-CD20) can trigger a transient IgM 'flare' (paradoxical rise) in the first weeks, worsening serum viscosity and bleeding risk. Therefore avoid single-agent rituximab when IgM is markedly elevated (often cited >40 g/L / >4000 mg/dL); plasmapheresis or holding rituximab until IgM is lowered prevents harm. Mnemonic: 'rituximab adds fuel to the fog before clearing it.'

BOARD TRAP

WRONG answer: giving upfront single-agent rituximab to a patient with very high IgM and hyperviscosity. The discriminator is the rituximab-induced IgM flare — first reduce IgM with plasmapheresis (or use a non-rituximab regimen), then add rituximab safely.